V6.2 · Comparison with V6

What improved, what is unchanged, and what it cost

Objective	State precisely how the V6.2 hierarchy differs from V6 on the same corpus and the same frozen inputs.
Method	Both stacks read the identical 24 frozen coordinates and the identical V6 MSS map M. Only the layer above MSS differs.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Figure 1 — V6 versus V6.2 dashboard.

What is identical

- The frozen atlas and its fingerprint 09ed804a40836f4a05a91ba10900cded.
- Preprocessing, the NNLS projection and the component registry.
- **The MSS layer.** M is read from V6 and never rewritten — the 17 motifs, their bands, exemplars, components and weights are byte-identical.

What changed

property	V6	V6.2
motif → theme map	hard partition (one theme per motif)	soft, row-stochastic, sparse
mean themes per motif	1.00	2.06
semantic levels	1 (K=13)	3 (K=17 / 6 / 4)
shared motifs	0 — suppressed by construction	4 represented
ontology	tree	graph, 11 multi-parent motifs
uncertainty	a scalar confidence	posterior, entropy, margin, propagated variance
theme output	a ranked class	a continuous non-negative coordinate
objective	κ × interpretability	4-objective Pareto (interpretability, information, recoverability, stability)
motif variance retained	0.981 (K=13)	0.751 (K=6)
compression	1.31×	2.83×

The cost, stated plainly. Compressing 17 motifs to 6 chemical themes retains 0.751 of the motif variance, against 0.981 at V6's K=13. V6.2 discards roughly 23 % more of the spectroscopic pattern in exchange for a level that is nameable, soft and multi-parent. Under the stated philosophy that is the intended trade; under a classification objective it would not be.

What was NOT re-measured

V6.2 changes the theme layer, so V6's per-theme top-1 numbers (theme top-1 0.614, top-3 0.890 at K=13) do not transfer to the new K=6 soft layer and are not restated here as if they did. The V6.2 analogue is the motif-level recoverability (0.503) plus the information and calibration measurements — deliberately, because analyte accuracy is the fourth priority, not the first.

Limitations

- The two hierarchies are not directly comparable on accuracy, because they abstract to different K and V6.2's output is continuous rather than a class.
- V6.2's L2 grouping is the chemical-superclass grouping, so part of the comparison is clustering-vs-chemistry rather than V6-vs-V6.2.
- Both share the same corpus limits: sterol, porphyrin, flavin and carotenoid remain under-grounded, and V6.2 inherits every one of them.

Recommendations

- Keep both layers available: V6's K=13 hard partition when information matters, V6.2's K=6 soft layer when interpretation and transfer matter.
- Do not report a V6.2 theme call without its entropy — the soft layer's value is that it can say 'both'.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from results/v6_rebuild/.